

fit@hcmus

Software Testing

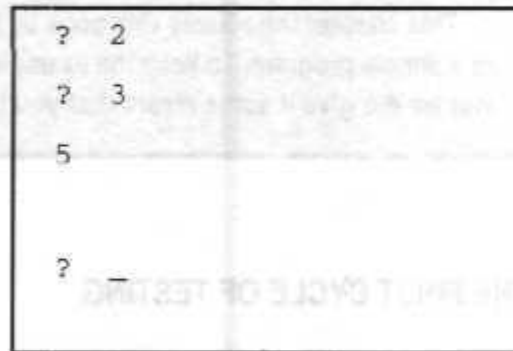
CSC13003

Software Testing Approaches

Exercise 1

- The program's specification
 - This program is designed to add 2 numbers, which you will enter
 - Each number should be one or two digits

Figure 1.2 How the screen looks after the first test



The cursor (beside the question mark at the bottom of the screen) shows you where the next number will be displayed.

Possible Test Cases

- Valid Cases: $199 \times 199 = 39,601$
 - $-99 \rightarrow -1$
 - $0 \rightarrow 99$
- Invalid Cases: INFINITE
 - ≤ -100
 - ≥ 100
 - Not a number

The problem

**very large or infinite
number of test scenarios
+
finite amount of time
=
impossible to test everything**

The solution

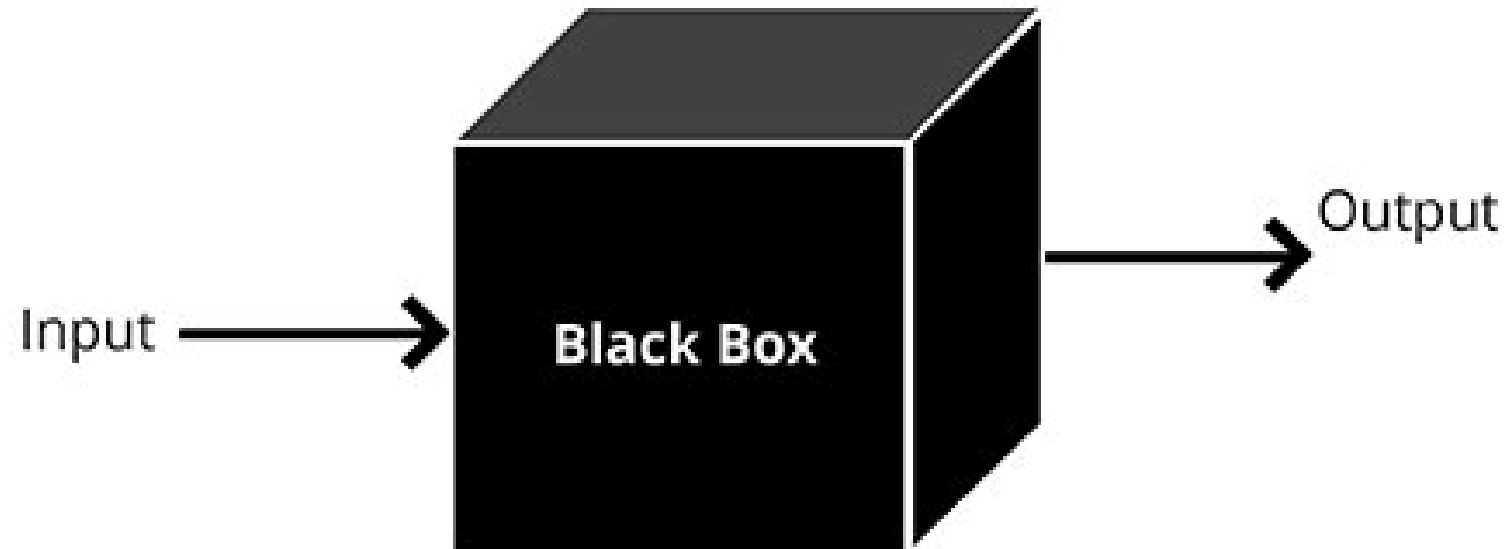
**Software testing strategies and methods (techniques)
exist to
reduce the number of tests to be run
whilst still providing sufficient coverage
of the system under test**

Testing Approaches



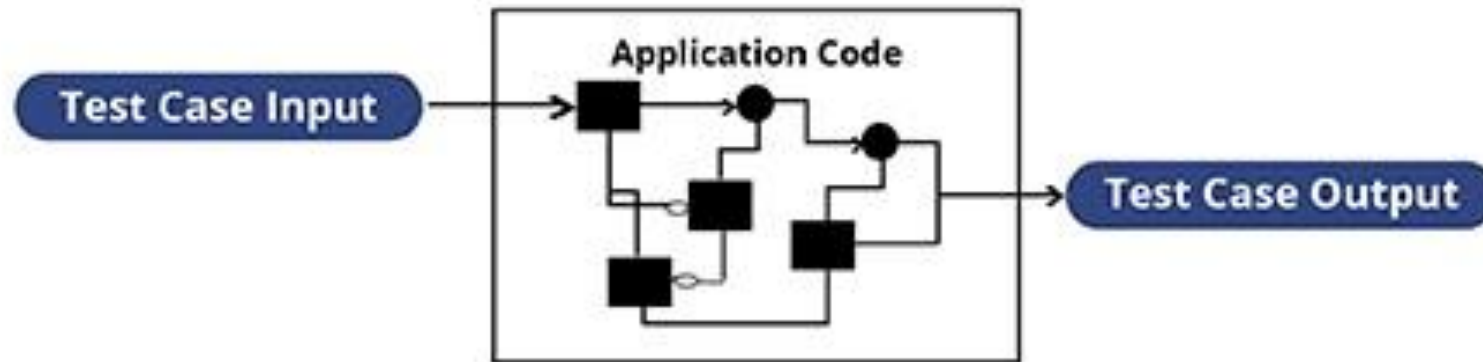
Black Box Testing Approach

BLACK BOX TESTING APPROACH

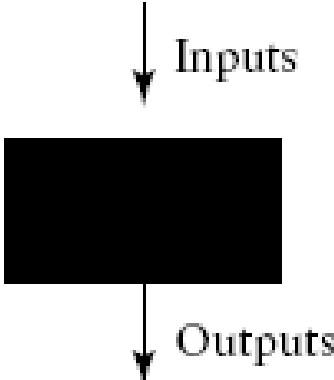
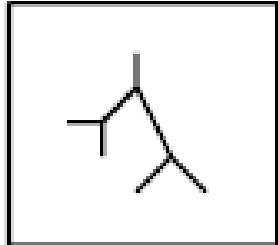


White Box Testing Approach

WHITE BOX TESTING APPROACH

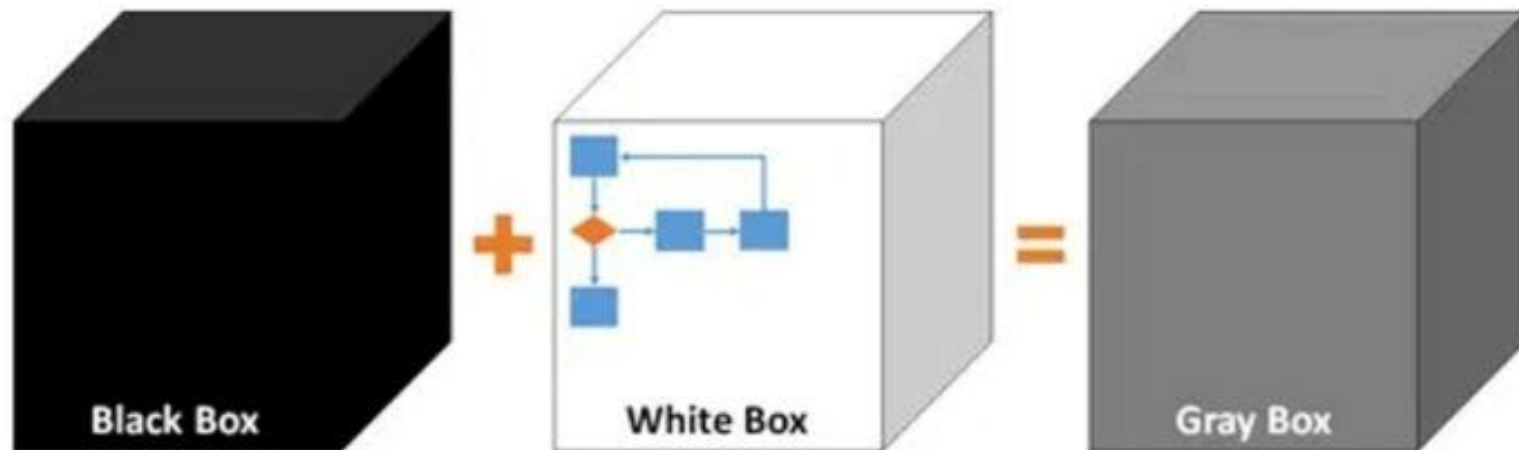


The two basic testing strategies

Test Strategy	Tester's View	Knowledge Sources	Methods
Black box		Requirements document Specifications Domain knowledge Defect analysis data	Equivalence class partitioning Boundary value analysis State transition testing Cause and effect graphing Error guessing
White box		High-level design Detailed design Control flow graphs Cyclomatic complexity	Statement testing Branch testing Path testing Data flow testing Mutation testing Loop testing

Grey Box Testing

Grey Box Testing



Black – Grey – White Box Testing

